

NGUYEN GIA HUY

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with hands-on experience building and operating ETL pipelines on AWS. Worked on ingestion, transformation, and batch processing for a large-scale utility platform using Lambda, ECS, S3, Athena, and RDS, and migrated its search workloads to OpenSearch. Currently builds account data migration tooling for a core banking platform.

TECHNICAL SKILLS

Languages: Python, SQL, Golang
Data Processing: ETL pipelines, batch processing, data migration
AWS: S3, Athena, Lambda, ECS, RDS
Storage & Streaming: PostgreSQL, OpenSearch, Elasticsearch, ClickHouse, Kafka
Infrastructure & Monitoring: Terraform, AWS SAM, CloudWatch, DataDog

WORK EXPERIENCE

Middle Software Engineer – Core Banking Data & Migration 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

- Built an account migration system in Golang on AWS (ECS, Lambda, RDS) for the bank's core banking platform, reducing migration time from **2.5 hours to 27 minutes**.
- Implemented interest accrual, penalty and late-fee calculation, repayment, and settlement for the Lending and Overdraft products, written as Python Smart Contracts on Thought Machine Vault Core.
- Connected Vault Core to the external backend services that support the bank's core operations and transaction workflows.
- Rebuilt the simulation testing framework in Golang, replacing a Gauge-based setup and reducing test runs from **24 minutes to just over 50 seconds**.
- Deployed a new Vault Core version on AWS (EKS, RDS, ECR, ELB) and set up its CI/CD pipeline.

Data Engineer (Software Engineer) 04/2025 – 03/2026
Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Designed, built, and operated ETL pipelines that ingest and transform large-scale utility data for au Denki's analytics and operational systems.
- Used Lambda for serverless processing steps and ECS for containerized batch jobs, with data stored in S3, queried through Athena, and RDS as the relational store.
- Wrote transformation logic that turns raw utility data into structured datasets for reporting and downstream systems.
- Optimized batch workflows by redesigning the pipelines, running independent jobs in parallel, and improving data access patterns, which increased throughput by **150%**.
- Migrated the search and analytics platform from Elasticsearch to OpenSearch and redesigned its ingestion and indexing flows, improving stability, data consistency, and query performance.
- Managed the platform's infrastructure with Terraform and deployed its serverless components with AWS SAM.
- Set up monitoring in CloudWatch and DataDog to track pipeline health, catch failures, and detect operational anomalies.

Software Engineer 04/2024 – 04/2025
VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Designed and operated a private cloud on OpenStack, using Cluster API to automate provisioning, scaling, and zero-downtime upgrades for **50+** production Kubernetes clusters.
- Implemented Keycloak-based multi-tenant authentication and authorization (RBAC) for **20+** customer organizations.

- Integrated and operated Kong Gateway for a microservices ecosystem of **100+** services, with service-level and network-level access control.
- Automated around **80%** of data center provisioning and infrastructure configuration tasks.

Blockchain Developer – On-chain Data Ingestion

06/2023 – 09/2023

EsolLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Built event-driven services that capture on-chain events, process them, and sync the results into off-chain databases and APIs.
- Integrated these services with **5+** blockchain networks, including Ethereum, BSC, Polygon, Aptos, and Sui, keeping transaction data consistent across chains.
- Developed and reviewed **15+** DeFi and NFT smart contracts in Solidity and Move.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built the backend with Kafka, MongoDB, Redis, and gRPC-Gateway, handling up to **1.5K** requests per second at 80–100 ms latency.
- Set up multi-cluster failover and observability with OpenTelemetry, Prometheus, Grafana, ELK, and ClickHouse; the platform ran at **99.95%** uptime.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on LiveKit with OpenAI-based transcription, voice translation, and meeting summaries.
- Designed the real-time translation pipeline and improved how conversation context is handled, which made translations more accurate and coherent.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. in Information Technology, Knowledge Engineering

DevOps on AWS Specialization | Coursera